KARTHIKEIN MR C72 BATCH

Assignment-based Subjective Questions:

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?
Seasonal Impact:

- Spring and Summer have more rentals of bikes due to good weather.
- Fall and Winter have less rentals because of cold weather.

Weather Condition:

- Sunny or overcast conditions increase bike rentals.
- Rainfall or heavy rainfall conditions decrease bike rentals.

Holiday vs Working Day:

- Rental is more during working days because of the necessity of commuting.
- Holiday has low rentals as it is a holiday and people either travel or tend to relax.

2. Why is it important to use drop_first=True during dummy variable creation?

The drop_first=True while creating the dummy variables is important for not falling under the dummy variable trap wherein the multicollinearity suffers in the model. Dropping one category would prevent perfect correlation between the dummy variables, and the model would be able to estimate the relationship between features and the target variable correctly.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

temp (temperature) has the strongest correlation with bike rentals. Typically, the more temperature increases, the more there is a tendency to increase in bike rentals.

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

I have checked the assumption by doing residual analysis that includes normality testing by histogram and Q-Q plots, also conducting Shapiro-Wilk test. For the model to be valid, residuals must follow a normal distribution.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

The top three features from the model are temp (temperature), hour, and weather situation, which also have a very high coefficient value in the model.

General Subjective Questions:

1. Explain the linear regression algorithm in detail.

It is a statistical method that models the relationship between the dependent variable and one or more independent variables, known as the predictors. In linear regression, the target variable is predicted based on the model by fitting a line that minimizes the sum of squared residuals. The aim is to determine the best fit line using the formula $y = mx + b$, where m is the slope and b is the intercept.

2. Explain the Anscombe's quartet in detail.

Analyze the Anscombe's quartet. It is four sets with nearly identical summary statistics and different distributions. It demonstrates a need to preview graphically before drawing conclusions. The datasets can show how two variables can have an equivalent correlation in the statistical realm but plot completely differently.

3. What is Pearson's R?

Pearson's correlation coefficient (r) measures the strength of the linear relationship between two variables. The values range between -1 to 1, -1 indicating a strong negative correlation, 1 being a strong positive correlation, and 0, no correlation.

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling is changing variables to the same scale. It is applied to avoid a bias in the model if variables have different units or magnitudes. Normalization scales data on a range from 0-1. The standardization has a mean and standard deviation scaled to 0 and 1, respectively.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

VIF (Variance Inflation Factor) tends to infinity in case of perfect multicollinearity between independent variables, where one variable can be perfectly predicted from others. This violates the assumption of independence in linear regression.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

A Q-Q plot is used to compare the distribution of residuals with a normal distribution. It helps in assessing the normality of residuals. If the residuals follow a straight line, the data is normally distributed, which is an assumption for linear regression.